

## Unit 4: Electricity

Electric current is the rate of flow of charge. The SI unit of current is the ampere (A).

1. A current of 2 A flows through a lamp for 30 s. Calculate the charge that passes.

Answer:  $Q = I \times t = 2 \text{ A} \times 30 \text{ s} = 60 \text{ C}$

2. State Ohm's law.

Answer: At constant temperature,  $V = I R$  for an ohmic conductor.

## Unit 5: Chemical reactions

3. Balance the equation:  $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$

Answer:  $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ . Mass is conserved:  $4 \text{ g} + 32 \text{ g} = 36 \text{ g}$ .

4. What is the pH of a neutral solution at 25 C?

Answer:  $\text{pH} = 7$ .

## Unit 6: Waves

The wave equation is  $v = f \times \lambda$ , where  $v$  is speed in m/s,  $f$  is frequency in Hz.

5. A sound wave has frequency 500 Hz and wavelength 0.68 m. Find its speed.

Answer:  $v = 500 \times 0.68 = 340 \text{ m/s}$

## ඒකකය 7: ජීර්කාශ සංශ්ලේෂණය

ජීර්කාශ සංශ්ලේෂණයේදී ශාක ආලෝක ශක්තිය භාවිතයෙන් ග්ලූකෝස් නිපදවයි.

6. ජීර්කාශ සංශ්ලේෂණයට අවශ්‍ය වායුව කුමක්ද?

පිළිතුර: කාබන් ඩයොක්සයිඩ් ( $\text{CO}_2$ )